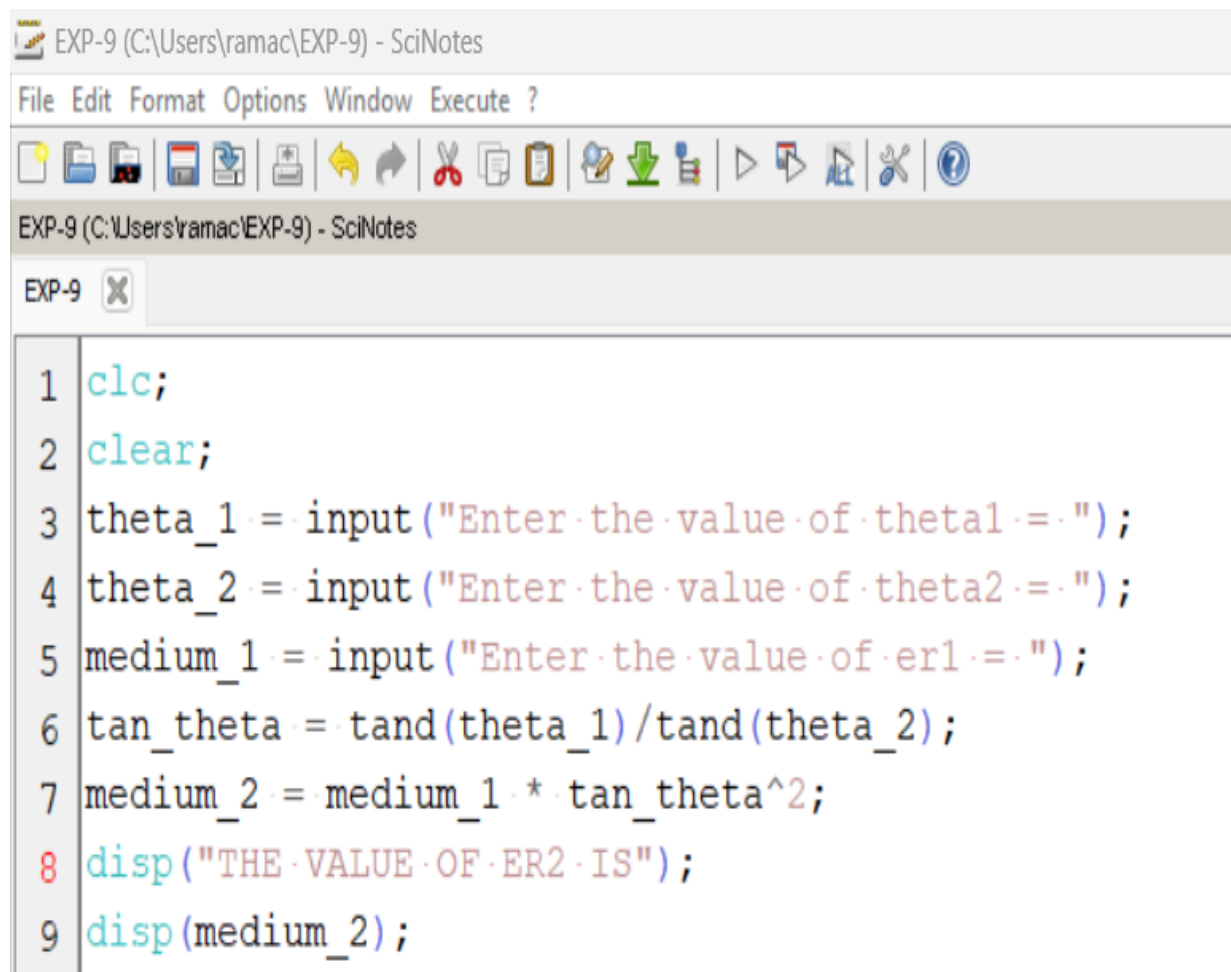


Exp:9 Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

Aim: To write program for Dielectric - free space boundary condition in Electric field using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



The screenshot displays the SCILAB 6.0.1 environment. The title bar indicates the file path 'EXP-9 (C:\Users\ramac\EXP-9) - SciNotes'. The menu bar includes 'File', 'Edit', 'Format', 'Options', 'Window', and 'Execute'. The toolbar contains various icons for file operations, editing, and execution. The main window shows the following SCILAB script:

```
1 clc;
2 clear;
3 theta_1 = input("Enter the value of theta1. = ");
4 theta_2 = input("Enter the value of theta2. = ");
5 medium_1 = input("Enter the value of er1. = ");
6 tan_theta = tand(theta_1)/tand(theta_2);
7 medium_2 = medium_1 * tan_theta^2;
8 disp("THE VALUE OF ER2 IS");
9 disp(medium_2);
```

```
Scilab 6.1.1 Console
File Edit Control Applications ?
[Icons]
Scilab 6.1.1 Console

Enter the value of thetal = 30

Enter the value of theta2 = 60

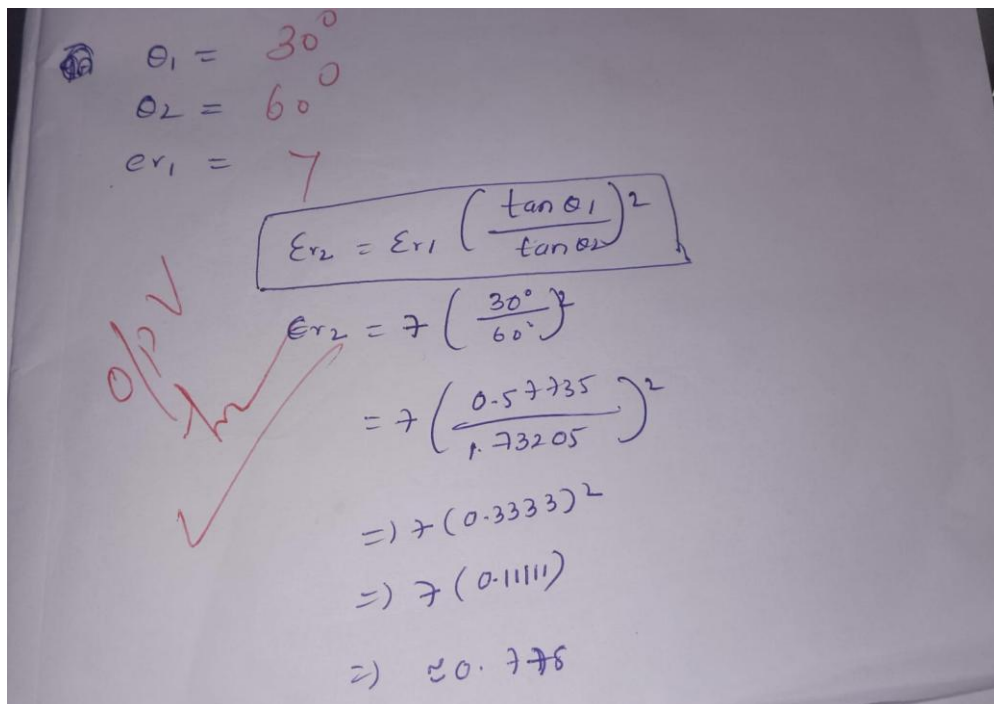
Enter the value of er1 = 7

"THE VALUE OF ER2 IS"

0.7777778

--> |
```

Theoretical Output:



Handwritten theoretical calculation for the dielectric constant ratio:

$$\begin{aligned}\theta_1 &= 30^\circ \\ \theta_2 &= 60^\circ \\ \epsilon_{r1} &= 7\end{aligned}$$
$$\boxed{\epsilon_{r2} = \epsilon_{r1} \left(\frac{\tan \theta_1}{\tan \theta_2} \right)^2}$$
$$\begin{aligned}\epsilon_{r2} &= 7 \left(\frac{30^\circ}{60^\circ} \right)^2 \\ &= 7 \left(\frac{0.57735}{1.73205} \right)^2 \\ &= 7 (0.3333)^2 \\ &= 7 (0.1111) \\ &= 0.7778\end{aligned}$$

The calculation is marked with a large red checkmark and the text "O/P ✓".

Result :

Thus the program was verified for Dielectric - free space boundary condition in Electric field using SCILAB was successfully.